

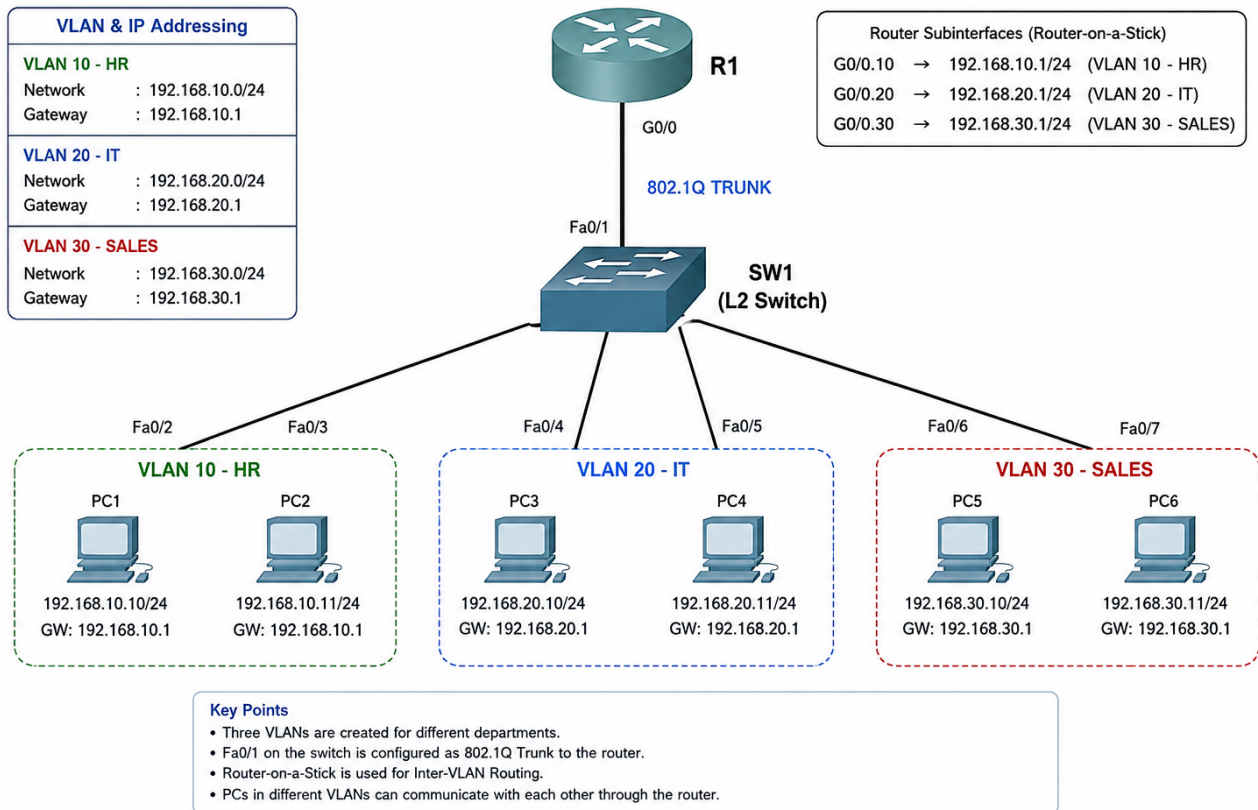
Small Office Network Design & Implementation

Cisco Packet Tracer | VLAN | 802.1Q Trunking | Router-on-a-Stick | Inter-VLAN Routing

1. Project Overview

This project demonstrates the design and implementation of a small office network using Cisco Packet Tracer. The network separates HR, IT, and Sales users into different VLANs and uses a Cisco router with subinterfaces to provide communication between the VLANs. The design demonstrates fundamental CCNA switching and routing concepts.

Small Office Network Topology



2. Objectives

- Create separate VLANs for HR, IT, and Sales departments.
- Assign IPv4 addresses and default gateways to end devices.
- Configure switch access ports for the appropriate VLANs.
- Configure an 802.1Q trunk between the switch and router.
- Implement Router-on-a-Stick using router subinterfaces.
- Enable communication between different VLANs.
- Verify the configuration using Cisco IOS commands and ICMP ping.

3. Network Topology

The topology contains one router (R1), one Layer-2 switch (SW1), and six PCs. The router-to-switch connection is configured as an 802.1Q trunk. Two PCs are assigned to each department VLAN.

4. VLAN & IP Addressing Plan

Department	VLAN	Network	Gateway	Example PCs
HR	10	192.168.10.0/24	192.168.10.1	192.168.10.10, .11
IT	20	192.168.20.0/24	192.168.20.1	192.168.20.10, .11
Sales	30	192.168.30.0/24	192.168.30.1	192.168.30.10, .11

5. Device & Port Mapping

Device/Link	Interface/Port	Purpose
R1	G0/0	Physical router interface
R1	G0/0.10	VLAN 10 gateway
R1	G0/0.20	VLAN 20 gateway
R1	G0/0.30	VLAN 30 gateway
SW1	Fa0/1	802.1Q trunk to R1
SW1	Fa0/2-3	HR access ports
SW1	Fa0/4-5	IT access ports
SW1	Fa0/6-7	Sales access ports

6. Switch Configuration

Create the department VLANs:

```
vlan 10 name HR vlan 20 name IT vlan 30 name SALES
```

Assign access ports:

```
interface range fa0/2-3 switchport mode access switchport access vlan 10 interface range fa0/4-5  
switchport mode access switchport access vlan 20 interface range fa0/6-7 switchport mode access  
switchport access vlan 30
```

Configure the trunk to the router:

```
interface fa0/1 switchport mode trunk no shutdown
```

7. Router Configuration — Router-on-a-Stick

Enable the physical interface and configure one subinterface for each VLAN:

```
interface g0/0 no shutdown interface g0/0.10 encapsulation dot1Q 10 ip address 192.168.10.1  
255.255.255.0 interface g0/0.20 encapsulation dot1Q 20 ip address 192.168.20.1 255.255.255.0 interface  
g0/0.30 encapsulation dot1Q 30 ip address 192.168.30.1 255.255.255.0
```

8. How Inter-VLAN Routing Works

A host in VLAN 10 cannot directly reach a host in VLAN 20 through Layer-2 switching because they belong to different IP networks. The source host sends the traffic to its default gateway (192.168.10.1). The switch carries the tagged traffic over the trunk to R1. R1 receives it on G0/0.10, routes it toward the 192.168.20.0/24 network, and sends it back through G0/0.20. The switch then forwards the traffic to the destination PC in VLAN 20.

9. Verification & Testing

Verify VLANs on the switch:

```
show vlan brief
```

Verify the trunk:

```
show interfaces trunk
```

Verify router interfaces:

```
show ip interface brief
```

Verify the routing table:

```
show ip route
```

Test inter-VLAN communication from a PC:

```
ping 192.168.20.10 ping 192.168.30.10
```

10. Expected Result

The switch should show VLANs 10, 20, and 30, with Fa0/1 operating as an 802.1Q trunk. R1 G0/0 and all three subinterfaces should be up/up. Hosts in different VLANs should successfully communicate through R1 when their IP address, subnet mask, VLAN assignment, and default gateway are correct.

11. Troubleshooting Checklist

- Check that G0/0 on R1 is not administratively down.
- Check that the switch-to-router port is trunking.

- Confirm VLAN IDs on the switch match the dot1Q IDs on the router.
- Confirm every PC is connected to the correct access VLAN.
- Confirm each PC has the correct default gateway.
- Use show vlan brief, show interfaces trunk, and show ip interface brief to isolate problems.

12. Skills Demonstrated

Cisco IOS CLI • VLAN segmentation • IPv4 addressing • Access ports • 802.1Q trunking • Router-on-a-Stick • Inter-VLAN Routing • Default gateways • ICMP troubleshooting • Packet Tracer network design

13. LinkedIn / GitHub Portfolio Use

Upload the .pkt file, this documentation PDF, the topology image, and verification screenshots to the GitHub repository. On LinkedIn, showcase the topology, VLAN configuration, trunk verification, router interface status, and successful inter-VLAN ping. Link the GitHub repository from the LinkedIn post so recruiters can inspect the complete technical work.